

Today's objective

① Review on vector & matrix derivatives

② Backpropagation

③ vector space

Partial Derivative for real value function

$$\text{let } f: \mathbb{R}^n \rightarrow \underbrace{\mathbb{R}}_{\text{real}}$$

vector gradient $f: x \in \mathbb{R}^n \rightarrow f(x)$

$$\underline{\nabla_x f} = \frac{df}{dx} = \left[\underbrace{\frac{\partial f(x)}{\partial x_1}, \dots, \frac{\partial f(x)}{\partial x_n}}_n \right]$$

example 1

$$f(x_1, x_2) = x_1^2 x_2 + x_1 x_2^3$$

$$\nabla_x f = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} & \frac{\partial f(x)}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2x_1 x_2 + x_2^3 & x_1^2 + 3x_1 x_2^2 \end{bmatrix}$$

example 2

$$f: \mathbb{R}^2 \rightarrow \mathbb{R} \quad x_1, x_2: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$\begin{bmatrix} s \\ t \end{bmatrix} \begin{matrix} \nearrow x_1(s, t) \\ \searrow x_2(s, t) \end{matrix} \rightarrow f(x_1, x_2)$$

$$\frac{df}{\underline{\partial(s, t)}} = \frac{\partial f}{\partial x} \cdot \underbrace{\begin{bmatrix} \frac{\partial x}{\partial s} \\ \frac{\partial x}{\partial t} \end{bmatrix}}_{\text{Jacobian}} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \end{bmatrix} \begin{bmatrix} \frac{\partial x_1}{\partial s} & \frac{\partial x_1}{\partial t} \\ \frac{\partial x_2}{\partial s} & \frac{\partial x_2}{\partial t} \end{bmatrix}$$

Partial derivative for vector value functions

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$x \in \mathbb{R}^n, \quad f(x) = \begin{bmatrix} \vec{f}_1(x) \\ \vdots \\ \vec{f}_m(x) \end{bmatrix} \in \mathbb{R}^m$$

$$\frac{\partial f}{\partial x_i} = \begin{bmatrix} \frac{\partial f_1}{\partial x_i} \\ \vdots \\ \frac{\partial f_m}{\partial x_i} \end{bmatrix}$$

Gradient of f : Jacobian Matrix J

$$J = \nabla_x f = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} & \dots & \frac{\partial f(x)}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(x)}{\partial x_1} & \dots & \frac{\partial f_m(x)}{\partial x_n} \end{bmatrix} \quad \left. \begin{array}{c} \\ \\ \end{array} \right\} m$$

→ Numerator Layout $\frac{df}{dx} \in \mathbb{R}^{m \times n}$

Alternative (denominator layout) $\frac{df}{dx} \in \mathbb{R}^{n \times m}$

Example Exercise $y = f(x)$

$$y_1 = -2x_1 + x_2$$

$$y_2 = x_1 + x_2$$

$$\text{compute } J = \nabla_x f = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix}$$

Example $f(x) = Ax$ $f(x) \in \mathbb{R}^m$, $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n$

$$\frac{df}{dx} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \dots & \frac{\partial f}{\partial x_n} \end{bmatrix}$$

Fact. $Ax = \sum_{j=1}^n x_j a_j$

$$\frac{\partial f}{\partial x_i} = \frac{\partial}{\partial x_i} Ax = \frac{\partial}{\partial x_i} \sum_{j=1}^n x_j a_j = \frac{\partial}{\partial x_i} x_i a_i = a_i$$

$$\frac{df}{dx} = \left[\frac{\partial f}{\partial x_1} \dots \frac{\partial f}{\partial x_n} \right] \quad \text{where } \frac{\partial f}{\partial x_i} = a_i$$

$$= [a_1 \dots a_n] = \underline{A}$$

This shows that $\frac{d}{dx} Ax = \underline{A}$.

(Approach 2) $Ax = \begin{bmatrix} \vec{a}_1^T x \\ \vdots \\ \vec{a}_n^T x \end{bmatrix} = \begin{bmatrix} \vec{a}_1^T x \\ \vdots \\ \vec{a}_n^T x \end{bmatrix}$

$$\frac{df}{dx} = \begin{bmatrix} \frac{df_1}{dx} \\ \vdots \\ \frac{df_n}{dx} \end{bmatrix} \rightarrow \frac{df_1}{dx} = \left[\frac{\partial}{\partial x_1} \vec{a}_1^T x \dots \frac{\partial}{\partial x_n} \vec{a}_1^T x \right]$$

$$= [a_{11}, \dots, a_{1n}] = \vec{a}_1^T$$

$$= \begin{bmatrix} \vec{a}_1^T \\ \vdots \\ \vec{a}_n^T \end{bmatrix} = \underline{A}$$

(Optional) Matrix-value function

$$f = Ax \in \mathbb{R}^m, \quad x \in \mathbb{R}^n, \quad A \in \underline{m \times n}$$

$$\nabla_A f = \frac{df}{dA} \in \mathbb{R}^{m \times (m \times n)}$$

$$= \begin{bmatrix} \frac{\partial f_1}{\partial A} \\ \vdots \\ \frac{\partial f_m}{\partial A} \end{bmatrix} \xrightarrow{\text{flatten}} (m \times n)$$

$$\frac{\partial f_i}{\partial A_{ij}} = \frac{\partial f_i}{\partial \vec{a}_i^T} = \frac{\partial}{\partial \vec{a}_i^T} Ax = x^T$$

$$\nabla_A f = I_p \otimes x^T = \text{diag}(x^T)$$

Useful Gradient Identities

① $a^T y \in \mathbb{R}^k$, $z = a^T y = \sum_{i=1}^k a_i y_i$

$$\frac{\partial z}{\partial y} = \begin{bmatrix} \frac{\partial z}{\partial y_1} \\ \vdots \\ \frac{\partial z}{\partial y_k} \end{bmatrix} = \begin{bmatrix} a_1 \\ \vdots \\ a_k \end{bmatrix}$$

② $z = y^T y = \sum_{i=1}^k y_i^2$

$$\frac{\partial z}{\partial y} = \begin{bmatrix} \frac{\partial z}{\partial y_1} \\ \vdots \\ \frac{\partial z}{\partial y_k} \end{bmatrix} = \begin{bmatrix} 2y_1 \\ \vdots \\ 2y_k \end{bmatrix} = 2y$$

③ $z = Ay$

$$\frac{\partial z}{\partial y} = A$$

④ $z = \underbrace{b^T A y}_{\in \mathbb{R}^n} = b^T \sum_{i=1}^k y_i a_i = \sum_{i=1}^k y_i \underbrace{(b^T a_i)}_{\in \mathbb{R}^n}$

$$\frac{d}{da} a^T b = b^T$$

$$\frac{\partial z}{\partial y} = \begin{bmatrix} \frac{\partial z}{\partial y_1} \\ \vdots \\ \frac{\partial z}{\partial y_k} \end{bmatrix} \quad \frac{\partial z}{\partial y_j} = \frac{\partial}{\partial y_j} (y_j (b^T a_j)) = a_j^T b$$

$$= \begin{bmatrix} a_1^T b \\ \vdots \\ a_k^T b \end{bmatrix} = \underbrace{A^T b}$$

$$\frac{d}{dy} Ay = A^T$$

⑤ $z = y^T A y = \sum_{i,j=1}^k y_i y_j a_{ij}$, find $\frac{\partial z}{\partial y_i} = 2y_i a_{i1}$

(hint)

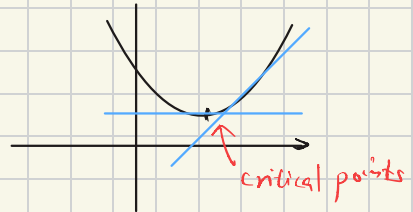
$$\frac{\partial z}{\partial y} = 2Ay$$

$$L(w) = \|y - Xw\|^2 \quad y, w \in \mathbb{R}^n \quad X \in n \times n$$

$$w^* = \arg \min_w L(w)$$

$\uparrow$ output $\uparrow$ input feature

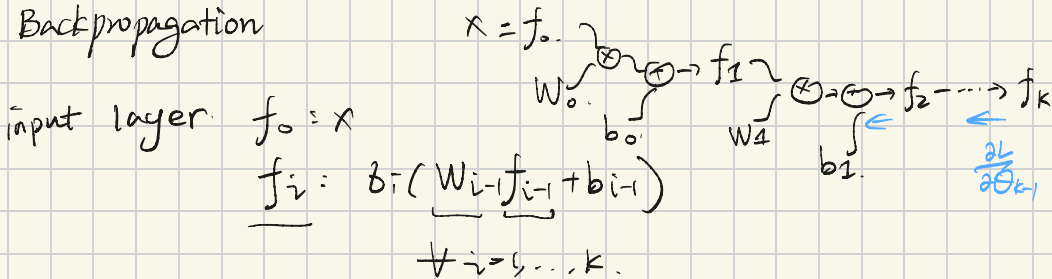
$$\nabla_w L(w) = 0 \quad \text{for critical point}$$



$L(w)$ is convex in $w \Rightarrow w^*$ is a global minimum if $\nabla_w L(w^*) = 0$.

$$\begin{aligned} \nabla_w L(w) &= (y - Xw)^T (y - Xw) \\ &= \nabla_w (y^T y - y^T Xw - (Xw)^T y + (Xw)^T Xw) \\ &= \nabla_w (-y^T Xw) - \nabla_w (Xw)^T y + \nabla_w w^T X^T Xw \\ &\stackrel{\text{by (5)}}{=} -X^T y \quad \underbrace{\nabla_w (y^T Xw)}_{-X^T y} \quad \stackrel{\text{by (6)}}{=} 2(X^T X)w \\ &= -X^T y + X^T X w = 0 \\ &\quad (X^T X)w = X^T y \\ &\quad w = (X^T X)^{-1} X^T y \end{aligned}$$

Backpropagation



$$L(\theta) = \|y - f_k(\theta; x)\|^2 \quad \text{where } \theta = \{w_0, b_0, w_1, b_1, \dots, w_k, b_k\}$$

$$\min_{\theta} L(\theta) \quad \text{via gradient descent}$$

$$\theta = \theta - \alpha \nabla_{\theta} L(\theta)$$

Apply chain Rule:

$$f_k \leftarrow \dots \leftarrow f_{k-1} \leftarrow f_k$$

$$\begin{aligned} \frac{\partial L}{\partial \theta_{k-1}} &= \frac{\partial L}{\partial f_k} \frac{\partial f_k}{\partial \theta_{k-1}} \quad \text{last layer.} \\ \frac{\partial L}{\partial \theta_{k-2}} &= \frac{\partial L}{\partial f_k} \frac{\partial f_k}{\partial \theta_{k-1}} \frac{\partial f_{k-1}}{\partial f_{k-2}} \\ &\vdots \\ \frac{\partial L}{\partial \theta_i} &= \left(\frac{\partial L}{\partial f_k} \dots \frac{\partial f_{i+2}}{\partial f_{i+1}} \right) \left(\frac{\partial f_{i+1}}{\partial \theta_i} \right) \\ &\quad \frac{\partial L}{\partial f_{i+1}} \end{aligned}$$